POWER BI_MODULE END PROJECT –TRANSPORT OPERATIONS DASHBOARD

Introduction

In this project, I created a Trip and Fuel Performance Dashboard using Power BI to analyze logistics operations.

Objective

To analyze transport operations data and identify key performance indicators such as delivery efficiency, fuel usage, maintenance cost, and operational cost per kilometer to optimize fleet performance and route planning.

Dataset Overview

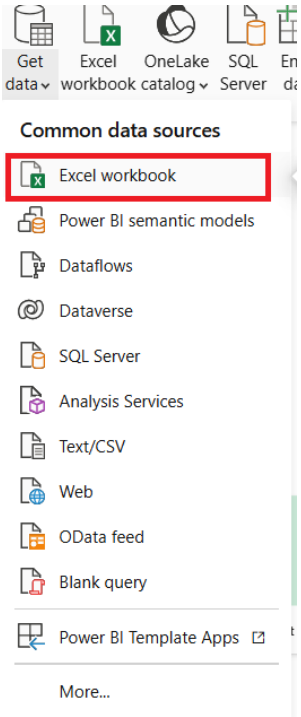
The dataset contains trip-based logistics data with 50 rows and 11 columns.

Main Tables Used:

1. **Trip_ID (Trip_Data)**
    - Trip_ID
    - Origin
    - Destination
    - Distance_km
    - Fuel_Consumed
    - Delivery_Status
    - Vehicle_ID
2. **Vehicle_ID Table**
    - Vehicle_Type
    - Vehicle_ID
    - Capacity_kg
    - Maintenance_Cost

Insert Dataset :

Power BI → Home → Get Data → **Excel Workbook**

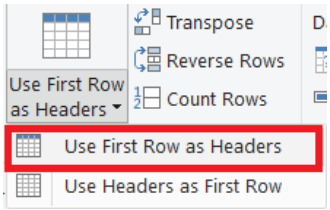


Selected sheets, and click Transform data to cleaning the dataset.

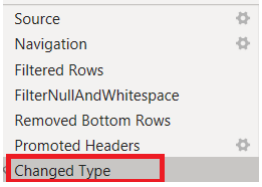
Power BI → Home → **Transform Data**

Data Cleaning & Modeling

1.Use first row as a Headers



2.Chaged Data type to entire column headers

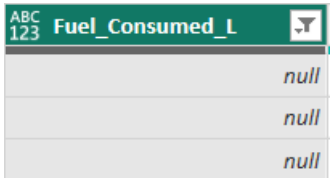


3.Handling Missing Values in Fuel_Consumed Column

Step 1: Identified Missing Values

While exploring the dataset in Power Query, I found that the Fuel_Consumed column contained 3 null (blank) values.

These missing values can affect total fuel calculation and fuel efficiency analysis.



Step 2: Checked Origin & Destination Combination

Since the dataset is trip-based, I analyzed the missing rows based on:

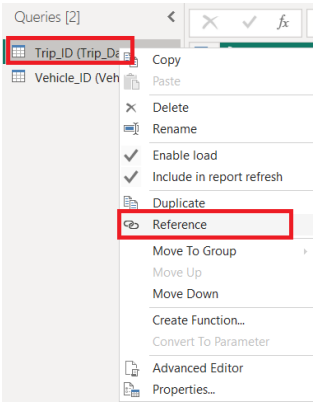
- Origin
- Destination

I checked whether similar route combinations were available in other rows.

Step 3: Created Reference Table

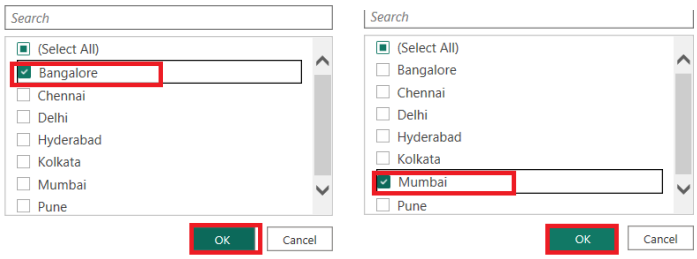
To calculate average fuel consumption by route:

1. I created a **Reference Table** from the original dataset.

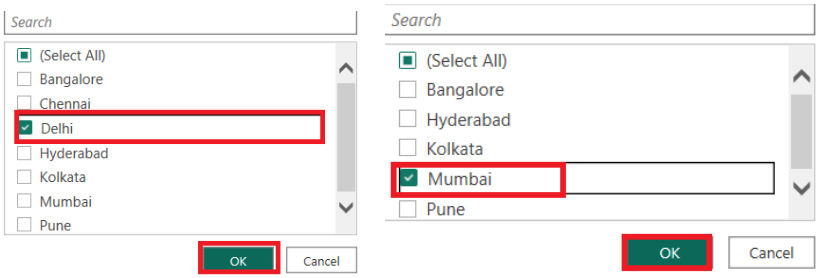


2. Grouped by:
   - Origin
   - Destination

   Bangalore → Mumbai (multiple rows available)



Delhi → Mumbai (only two rows)

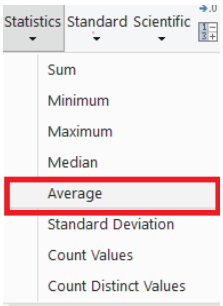


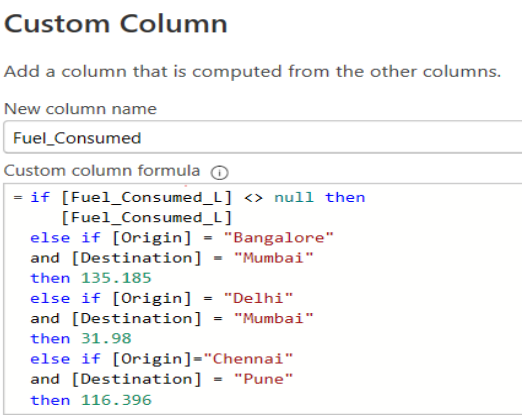
Chennai → Pune (only one row)

1		D04	Chennai	Pune

3.Calculated:

- Average of Fuel_Consumed





I applied hierarchical averaging — starting with route-level, then origin-level, then overall average to preserve data integrity.

Step 4: Replaced Null Values

Used Conditional Column:

If Fuel_Consumed is null
→ Replace with Route Average Fuel

Else
→ Keep original value

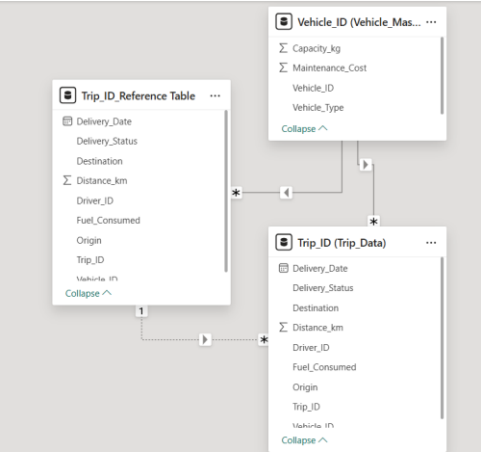
This ensured:

- No duplicate columns created
- Single clean Fuel_Consumed column maintained

Step 5: Final Validation

- Verified no null values remain in Fuel_Consumed
- Checked totals before and after replacement
- Ensured no impact on data consistency

Data Modeling :



- Created relationship between Trip_Data and Vehicle_ID table
- Relationship type: One-to-Many (*:1)
- Ensured no blank keys

DAX Measures:

Fuel Efficiency = Distance / Fuel Consumed

```
Fuel_Efficiency = DIVIDE(SUM('Trip_ID (Trip_Data)'[Distance_km]),SUM('Trip_ID (Trip_Data)'[Fuel_Consumed]))
```

Fuel efficiency was calculated to evaluate vehicle performance and control fuel-related operational costs, which directly impact overall cost per kilometer.

On-Time Delivery % = On-Time Trips / Total Trips

```
On Time %_Delivery =
DIVIDE(
    CALCULATE(
        DISTINCTCOUNT('Trip_ID (Trip_Data)'[Trip_ID]),
        'Trip_ID (Trip_Data)'[Delivery_Status] = "On-Time"
    ),
    DISTINCTCOUNT('Trip_ID (Trip_Data)'[Trip_ID])
)
```

On-Time Delivery percentage was calculated to evaluate service reliability and customer satisfaction, and to identify performance gaps across drivers and destinations.

Cost per km = (fuel cost + Maintenance Cost) / Distance

```
Fuel_Cost = 'Trip_ID (Trip_Data)'[Fuel_Consumed] * 100
```

```
Cost_Per_KM = DIVIDE(SUM('Trip_ID (Trip_Data)'[Fuel_Cost]) + SUM('Vehicle_ID (Vehicle_Master)'[Maintenance_Cost]),SUM('Trip_ID (Trip_Data)'[Distance_km]))
```

Cost per kilometer was calculated to evaluate operational efficiency and monitor fleet operating expenses, as it directly impacts profitability and route optimization decisions.

<u>Cards Visualizations :</u>

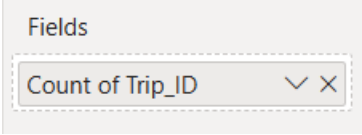
I created KPI cards at the top of the dashboard to provide a quick summary of overall performance.

The following KPIs were included:

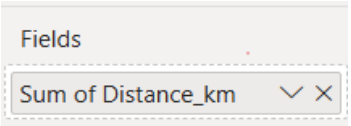
- **Average Cost per KM**

```
Cost_Per_KM = DIVIDE(SUM('Trip_ID (Trip_Data)'[Fuel_Cost]) + SUM('Vehicle_ID (Vehicle_Master)'[Maintenance_Cost]),SUM('Trip_ID (Trip_Data)'[Distance_km]))
```

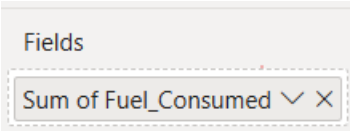
- **Total Trips**



- **Total Distance Covered**



- **Total Fuel Consumed**



- **On-Time Delivery %**

```
On Time % =
DIVIDE(
    CALCULATE(
        DISTINCTCOUNT('Trip_ID (Trip_Data)'[Trip_ID]),
        'Trip_ID (Trip_Data)'[Delivery_Status] = "On-Time"
    ),
    DISTINCTCOUNT('Trip_ID (Trip_Data)'[Trip_ID])
)
```

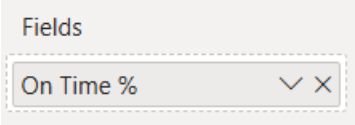
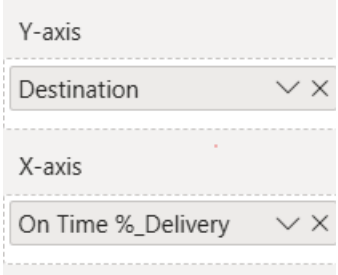
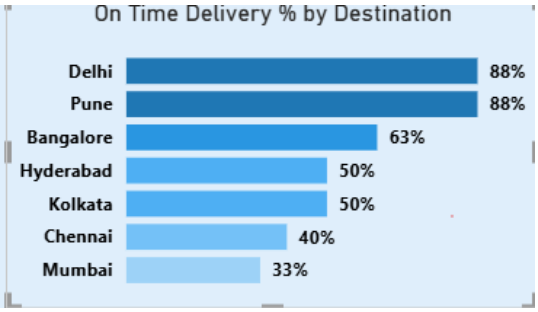


Chart Visualizations:

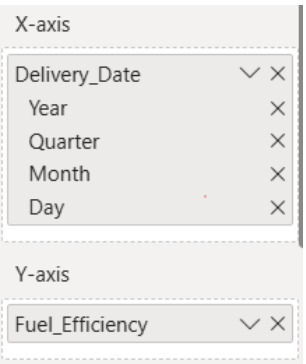
Bar chart: On-Time Delivery % by Destination

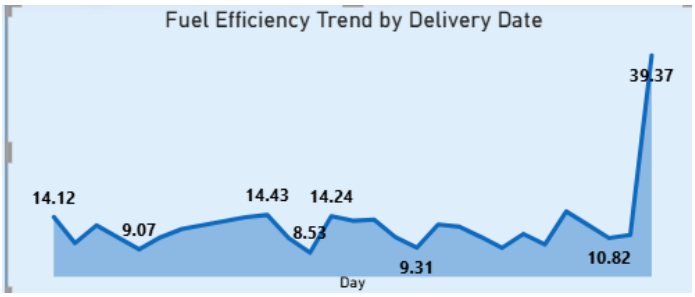




On-Time Delivery percentage by destination was analyzed to identify region-wise service performance gaps and optimize route planning and resource allocation.

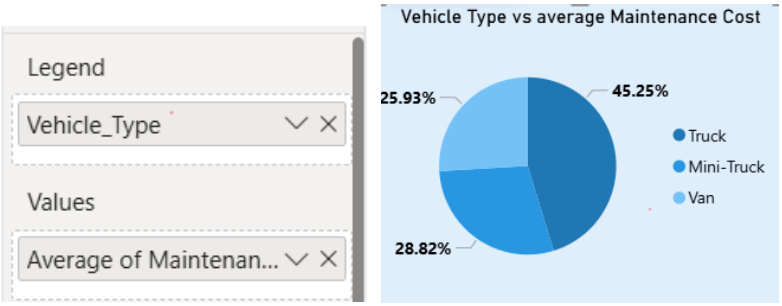
Line chart: Fuel efficiency trend by delivery date.





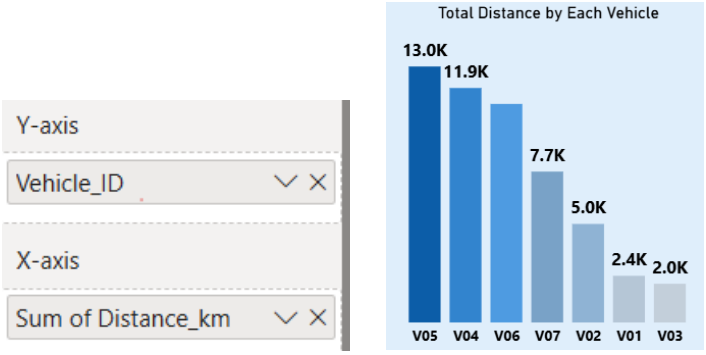
Fuel Efficiency trend by delivery date was analyzed to monitor vehicle performance over time, detect performance decline early, and support preventive maintenance and fuel cost optimization**.**

Pie chart: vehicle type vs Average maintenance cost



This analysis helps identify which vehicle category incurs higher maintenance expenses, enabling better fleet planning, cost optimization, and long-term investment decisions.

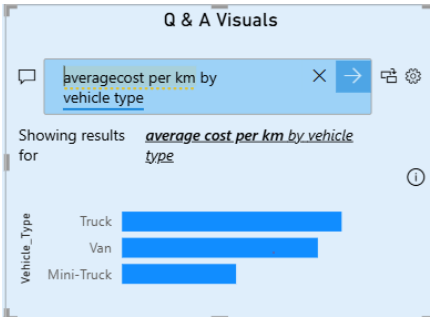
Column chart: Total distance by Each vehicle



Total distance by vehicle was analyzed to evaluate fleet utilization, balance workload, and support maintenance planning and cost optimization decisions.

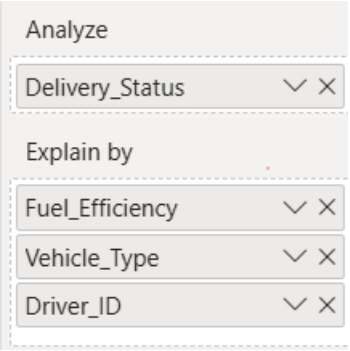
<u>AI-Powered Visuals:</u>

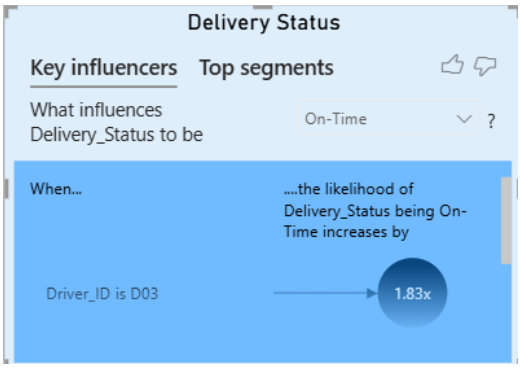
Q&A Visual:



Q&A visual was included to enable interactive, natural-language-based data exploration, allowing business users to quickly generate insights without requiring technical knowledge of Power BI.

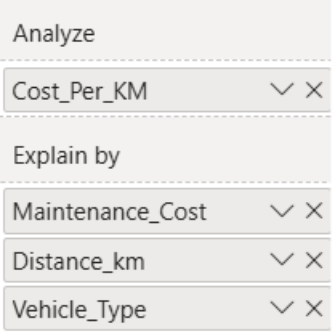
Key Influencers Visual:

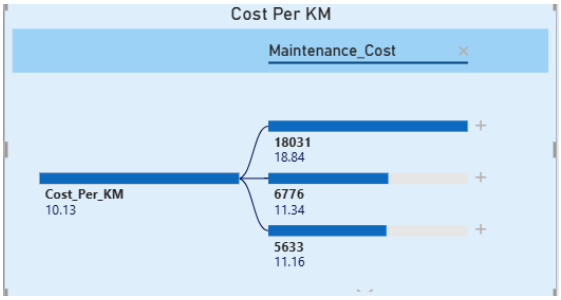




The Key Influencers visual was used to identify the major factors affecting delivery performance, enabling root cause analysis and data-driven operational improvements.

Decomposition Tree (AI Visual):





The Decomposition Tree was used to break down Cost per km across multiple dimensions to identify key cost drivers and support strategic fleet cost optimization.

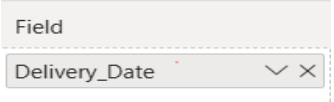
<u>Slicers (Filters)</u>

I added slicers for:

- **Origin**

- **Destination**

- **Date**

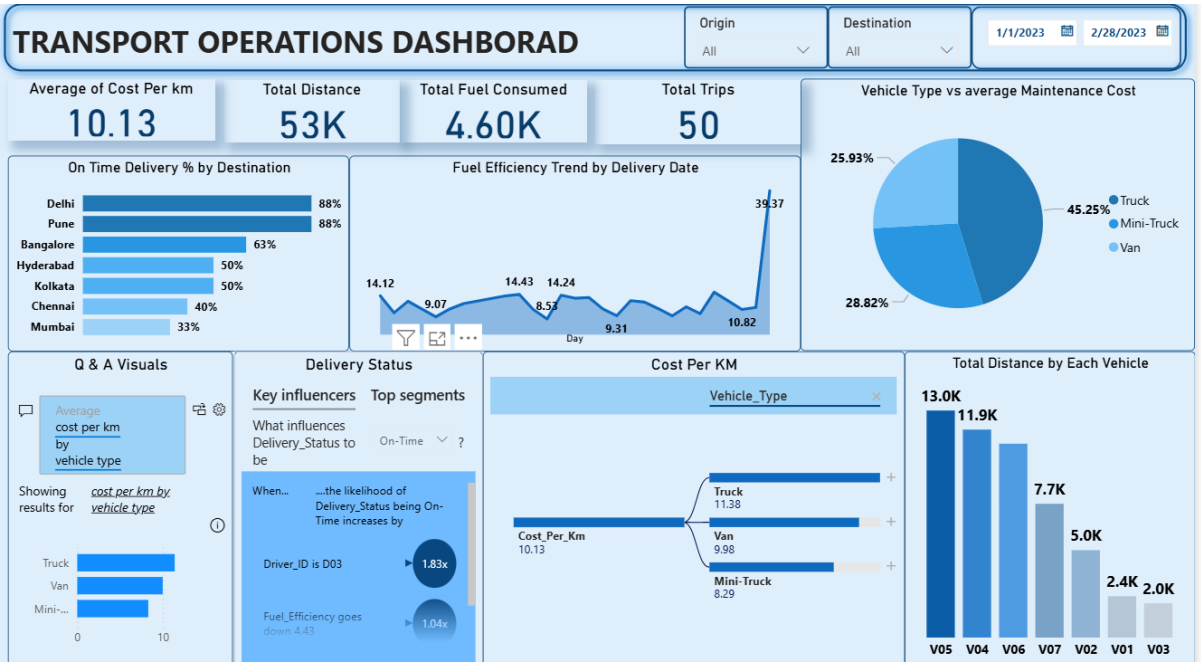


These slicers allow users to dynamically filter and interact with the dashboard.This improves usability and interactivity.

Key Insights :

1.Cost efficiency depends heavily on vehicle type.
2. Truck operations increase cost per km.
3. Fuel efficiency instability impacts cost & delays.
4. On-time performance varies significantly by destination.
5. Driver performance influences service reliability.
6. Fleet utilization is imbalanced.

Conclusion :

The Transport Operations Dashboard reveals that while the fleet operates at an average cost of ₹10.13 per km, operational efficiency varies significantly across vehicle types and destinations. Trucks contribute the highest cost per km and maintenance expenses, while mini-trucks offer better cost efficiency. On-time delivery performance is strong in Delhi and Pune but weak in Mumbai and Chennai, indicating route-level inefficiencies. Fuel efficiency fluctuations and uneven vehicle utilization suggest the need for preventive maintenance and better fleet allocation. Overall, optimizing vehicle mix, improving driver performance, and balancing route assignments can significantly enhance profitability and service reliability.

Final Dashboard View :